Dynamic Model Fusion for Multi-Source Test-Time Adaptation

Yuan Xue[a], Qinting Jiang[a], Yuan Meng[a], Xingxuan Zhang[a], Chen Tang[b], Jingyan Jiang[c] and Zhi Wang[a,*]

[a]Tsinghua University
[b]The Chinese University of Hong Kong
[c]Shenzhen Technology University

Abstract. Deep Neural Networks suffer significant performance degradation when faced with distribution shifts between training and test data. Test-time adaptation (TTA) has emerged as a practical solution that enables models to adapt to the shifted test distribution. Currently, most existing TTA methods are designed around a single model, which incorporate limited information from a singular data distribution. In practice, pre-trained models derived from diverse source domains are readily accessible, each capturing distinct data distribution and containing complementary information. To exploit this diversity, we propose Model Fusion-based multi-source Test-Time Adaptation (MFTTA), which constructs a target model by fusing the parameters of multiple source models. Drawing inspiration from deep model fusion, we introduce a fine-grained fusion mechanism governed by an off-policy reinforcement learning agent, which dynamically assigns fusion weights based on the current data distribution. Furthermore, we design a correlation-aware model update strategy that prioritizes the source model most relevant to the incoming test data. Extensive experiments on standard out-of-distribution benchmarks demonstrate that our method effectively integrates knowledge from multiple source models, adapts robustly to dynamic distribution shifts, and alleviates the problem of forgetting in long-term adaptation.

1 Introduction

Deep Neural Networks (DNNs) have demonstrated remarkable performance in various challenging tasks, including image classification, object detection, and natural language processing. This success is largely predicated on the assumption that the distribution of the test data closely matches that of the training data [34, 40, 41]. In real-world deployments, this assumption often fails due to various types of distribution shifts. These shifts arise from dynamic environmental factors, such as changes in weather conditions (e.g., rain, snow, fog), temporal variations (e.g., morning, noon, night), or sensor-related degradations (e.g., Gaussian noise, defocus blur) that compromise image quality [29]. Even subtle deviations from the original data distribution can cause significant drops in model accuracy, thereby limiting the robustness and reliability of pre-trained models in uncontrolled settings. Addressing these challenges is critical to unlocking the full potential of DNNs in safety-critical and production-grade applications.

* Corresponding Author. Email: wangzhi@sz.tsinghua.edu.cn

To mitigate the adverse effects of distribution shifts, unsupervised domain adaptation has been extensively studied. These approaches aim to align feature distributions between the labeled source domain and the unlabeled target domain, enabling improved generalization on the latter [29, 41]. However, such methods typically assume simultaneous access to source and target data, which is infeasible due to privacy concerns, storage constraints, or changing target environments. To address these limitations, test-time adaptation (TTA) has emerged as a promising alternative that adapts models online using only unlabeled test data. Representative methods such as Tent [39] optimize the affine parameters in batch normalization (BN) layers during inference, while CoTTA [42] leverages a combination of temporal and augmentation-based ensembling to stabilize predictions. These methods avoid full retraining and thus offer a computationally efficient means to enhance robustness against distribution shifts, especially in streaming or real-time inference settings.

Previous research on TTA has concentrated on the online adaptation of a single-source model [39, 11, 42, 26, 27, 15]. In practice, multiple pre-trained models are often accessible, which are optimized on different source domains. These models may capture complementary information and exhibit varying levels of performance depending on the characteristics of the current target data [23, 44]. For example, consider models trained on sunny and rainy weather data. A target domain characterized by overcast conditions may not align perfectly with either source domain but can benefit from the knowledge embedded in both. Relying on a single source model in such cases may lead to suboptimal performance, whereas a collaborative strategy that draws upon multiple models could offer stronger generalization capabilities [4]. Consequently, leveraging multiple source models in a unified manner represents a valuable yet underexplored direction for TTA.

The primary challenge in multi-source TTA lies in designing an effective collaboration strategy that maximizes performance gains while minimizing resource overhead. Most existing approaches draw inspiration from ensemble learning, which assumes that model knowledge is encoded in its output. In this setting, multiple independently trained source models contribute to the final prediction through result aggregation, such as weighted voting or averaging [4, 44]. Although effective in improving prediction reliability, ensemble learning suffers from substantial computational costs, especially during inference. Taking a standard ResNet-50 model as an instance, which contains approximately 25.6 million parameters and requires over 4.1 billion floating-point operations (FLOPs) per

forward pass. When multiple such models are involved in ensemble prediction, the cumulative cost becomes prohibitive—particularly on resource-constrained devices. In contrast, deep model fusion (DMF) offers a more efficient alternative: by interpolating the parameters of multiple source models to construct a single target model, inference can be conducted once on the fused model rather than multiple times on each source model [2, 24]. This "parameter interpolation followed by single inference" paradigm is significantly more cost-effective than "multiple inferences followed by result interpolation". Motivated by these considerations, we advocate for a model fusion approach to multi-source test-time adaptation that preserves the strengths of model collaboration while achieving superior computational efficiency.

Building upon the DMF concept, we introduce Model Fusion-based multi-source Test-Time Adaptation (MFTTA), a novel framework that dynamically generates target models tailored to the current test-time distribution. Our method employs an off-policy reinforcement learning (RL) algorithm for weight decision, which adaptively determines the optimal fusion weights for source models. This design enables the system to respond intelligently to shifting data characteristics, generating a target model that generalizes effectively without access to labeled data. Additionally, we introduce a correlation-based update mechanism that prioritizes the most relevant source model at each step, enhancing adaptation efficiency by focusing learning efforts on the most informative sources. The main contributions of this paper are summarized as follows:

- We introduce DMF as a novel collaboration strategy for multi-source test-time adaptation. Unlike traditional ensemble learning methods that operate at the output level, DMF integrates multiple source models by directly aggregating their parameters. This approach allows us to construct a single target model that inherits diverse knowledge from multiple domains.
- We first investigate the fine-grained model fusion. Specifically, we analyze how fusion weights influence the generalization ability of the fused model. Building on this insight, we propose an RL-based model fusion strategy that dynamically determines fusion weights at test time, enabling the model to adjust continuously to evolving data distributions.
- We validate the effectiveness of our proposed method through extensive experiments on widely-used out-of-distribution (OOD) benchmarks. Our results show that MFTTA not only harnesses the complementary strengths of individual source models, but also mitigates the problem of catastrophic forgetting during prolonged test-time adaptation.

2 Related Work

Test-Time Adaptation. Given a well-trained classifier on the source domain and unlabeled target domain, source-free domain adaptation (SFDA) aims to leverage the implicit source domain knowledge within the classifier to perform inference on all samples in the target domain, which requires all test samples to be available [21, 25, 20, 22]. DECISION [1] addresses the problem of unsupervised multi-source domain adaptation without access to source data by proposing an algorithm based on pseudo-labeling and the principle of information maximization. Shen et al. [33] investigate theoretical bounds on the generalization error of the resulting target model from an information-theoretic perspective, and achieve a trade-off between bias and variance through domain aggregation, selective pseudo-labeling, and joint feature alignment. EAAF [28] introduces an unsupervised multi-source adaptation method based on evidential learning, aiming to optimize the utilization efficiency of each source domain through a more fine-grained aggregation strategy. Compared to SFDA, online test-time adaptation only has access to the current batch (without having the complete target domain data) for adaptation [21]. Wang et al. [39] utilized the entropy minimization of pseudo-labels to update the affine parameters of BN layers. CoTTA [42] is the first dedicated method proposed for continuous testing time adaptation, which mitigates error accumulation issues by employing weighted averaging and enhanced average prediction. Niu et al. [26] proposed a reliable sample-based TTA method called EATA, which utilizes sample entropy measurement to select reliable pseudo-labels for model adaptation and applies Fisher regularization to alleviate catastrophic forgetting. Although online TTA no longer relies on complete target domain data, existing TTA methods only focus on the single-source setting and cannot harness the potential of collaborative adaptation for multi-source scenarios.

Deep Model Fusion. DMF methods can be categorized into training phase fusion and test phase fusion. In the training phase, models may get trapped in loss landscape valleys and converge to local optima [16, 5]. Izmailov et al. [16] propose collecting model weights at different time points during the iteration process and averaging these weights. Fusing [7] fine-tunes the same pre-trained model on different auxiliary datasets and then merges the resulting fine-tuned models into a single one, followed by a final fine-tuning step on the target task. Model Ratatouille [31] further delays the fusion process: it first performs a second-stage fine-tuning on each model using the target dataset to reduce weight conflicts, and then conducts model fusion. To achieve model fusion in the testing phase, alignment of parameters between the source models needs to be accomplished [2, 43, 24, 17, 36, 35]. Ainsworth et al. [2] introduced weight-based and activation-based matching methods to compute permutation matrices for each layer of the source models. Parameter interpolation is then performed on the permuted source models, constructing a target model that inherits the recognition capabilities of the source models. Existing model fusion methods only consider static fusion, which constructs a target model for the fixed target domain. However, when the target domain continues to change, static fusion is unable to handle all situations. Therefore, it is necessary to dynamically adjust fusion parameters for the changing data stream.

3 Measurement & Motivation

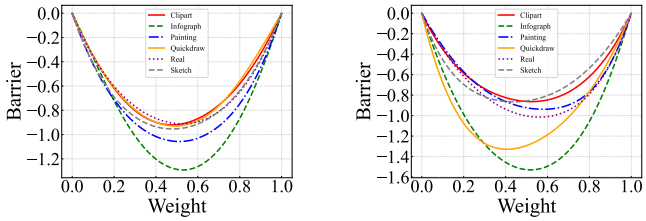
3.1 Preliminaries

In the loss landscape of DNN training, the minimum loss is typically described as a point at the bottom of a convex valley, and the model parameters are determined by the location of this minimum [6, 18]. The conventional view holds that the loss landscape contains numerous local minima and saddle points, with different local minima corresponding to potentially isolated regions where the model may converge [18, 2, 9]. As a result, a linear path directly connecting two independent models generally forms a high loss barrier, which is calculated as:

$$B(\theta_1, \theta_2) = \sup_t \left[\mathcal{L}(w\theta_1 + (1-w)\theta_2)\right] - \left[w\mathcal{L}(\theta_1) + (1-w)\mathcal{L}(\theta_2)\right], \quad (1)$$

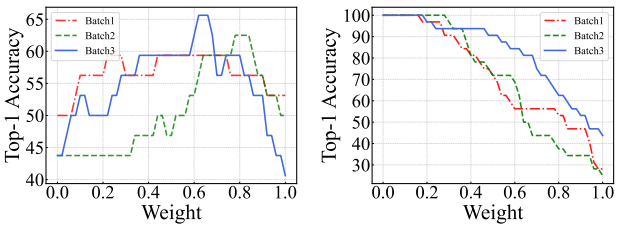
where θ_1, θ_2 represent parameters of two models respectively, w is the linear interpolation weight, and $\mathcal{L}$ denotes the loss function.

The loss barrier represents the variation in loss along the optimization path between θ_1 and θ_2. If there exists a path connecting the two models with a near-zero loss barrier, it indicates the presence of mode



(a) Clipart+Infograph (b) Painting+Sketch

Figure 1: Loss barrier of source model pairs. The upper bound of the loss barrier being zero across all target domains indicates the validity of mode connectivity, suggesting that model fusion has the potential to deliver better performance than ensemble learning.



(a) Clipart (b) Painting

Figure 2: Accuracy-weight curves for target domain Clipart and Painting. The source domains are Painting and Sketch. For each target domain, 3 batches are randomly selected. The optimal weights vary for different batches, and the selection of weights significantly impacts the accuracy.

connectivity between them [9]. In this case, a target solution θ_t can be constructed along the path by fusing the two models. Extending to the case with multiple models, the target model parameter can be constructed as:

$$\theta_t = \sum_{s=1}^{S} \lambda_s * \mathcal{P}(\theta_s), \tag{2}$$

where θ_s denotes the parameter of source model M_s, and λ_s denotes the fusion weight of M_s. $\mathcal{P}$ represents the permutation operation in DMF, which is a model transformation designed to construct a near-zero loss barrier [2, 35].

3.2 *Deep Model Fusion of Source Models*

Following the above description, we fine-tune source models from pre-trained ResNet-50 on each of the six domains from DomainNet [30] and pair them to calculate the loss barrier. $\mathcal{P}$ is set as the identity matrix, indicating that no parameter permutation is performed. Fig. 1 presents the experimental results of how the loss barrier varies with weight, where the weights are distributed as fifty equally spaced values between 0 and 1. The loss barrier consistently approaches zero, suggesting intrinsic mode connectivity between the source models without extra permutation operations. In addition, the loss barriers are negative at all weights, indicating that the first term in Eq. 1 is less than the second term (i.e., the loss of interpolation is less than the interpolation of loss). This suggests that the fused model may outperform the ensemble of source models.

We then investigate the sensitivity of different batches to λ_s and the variability of optimal weights. Fig. 2 displays the classification accuracy of three batches with the weights divided at a granularity of 0.02. Taking Fig. 2a as an example, the range of accuracy for the first batch (red) is 9.375%, with the highest accuracy corresponding to weight ranges of [0.22, 0.78]-[0.28, 0.72] and [0.44, 0.56]-[0.74, 0.26]; for the second batch (green), the range of accuracy is 18.75%, with the highest accuracy weight range being [0.78,0.22]-[0.84, 0.16]; for the third batch (blue), the range of accuracy is 25%, with the highest accuracy weight range at [0.62, 0.38]-[0.66, 0.34]. The experiments demonstrate significant variations in the highest accuracy across different batches, with the optimal weights also exhibiting variability. For OOD target domains, the accuracy of the target model tends to be higher than that of the source models. Conversely, for in-distribution (ID) target domains, the proportion of the corresponding source domain is generally higher.

Table 1: Top-1 accuracy of Leave-one-domain-out experiments. C, I, P, Q, R and S are the abbreviations of Clipart, Infograph, Painting, Quickdraw, Real and Sketch in DomainNet respectively. Source Best refers to inferring the target domain using all source models and selecting the best result. Results marked with † and ‡ are reported by Gulrajani et al. [12] and Cha et al. [5], respectively.

Method	C	I	P	Q	R	S	Avg
IRM†[3]	48.5	15.0	38.3	10.9	48.2	42.3	33.9
MLDG†[19]	59.1	19.1	45.8	13.4	59.6	50.2	41.2
CORAL†[37]	59.2	19.7	46.6	13.4	59.8	50.1	41.5
ERM‡[38]	62.8	20.2	50.3	13.7	63.7	52.1	43.8
SAM‡[10]	64.5	20.7	50.2	15.1	62.6	52.7	44.3
EOA [4]	65.9	23.4	55.3	16.5	66.4	57.1	47.4
Source Best	49.6	17.4	46.4	11.6	50.2	41.9	36.2
Ensemble - Avg	60.2	17.7	46.2	11.6	59.8	50.7	41.0
- Best	60.5	19.9	51.9	13.8	62.0	51.7	43.3
- Batch-wise	70.8	26.9	60.8	23.9	73.5	61.1	52.8
- Sample-wise	80.4	32.1	69.1	30.4	81.4	71.1	60.7
Lerp - Avg	40.2	11.1	19.3	7.6	47.2	27.7	25.5
- Best	54.7	18.1	46.4	13.8	58.1	47.8	39.8
- Batch-wise	69.4	27.6	59.8	26.2	74.7	61.2	53.2
- Sample-wise	85.1	36.3	74.2	35.4	85.2	76.2	65.4

Next, we measure the performance upper bound of fine-grained model fusion. To mitigate the impact of source domain selection, we employ the Leave-one-domain-out [12] setting to assess the generalization performance of the model fusion method. Specifically, in each iteration, one domain is chosen as the unknown target domain, with all other domains serving as known source domains. Additionally, we select other comparative domain generalization (DG) methods, with results shown in Table 1. For source models, Table 1 compares the performance of ensemble learning and model fusion, where "Avg" indicates that the weight of each source domain is $1/S$, and "Best" represents the optimal result in the weight space. We use linear interpolation as the fusion method, named "Lerp". Experimental results indicate that, under the "Avg" setting, there is still a significant gap between the performance of model fusion and ensemble, while the accuracy of model fusion is only 3.5% lower than that of ensemble under the "Best" setting.

In the context of fine-grained model fusion, we first define a set of candidate weight $\lambda_{cand}(\varepsilon) = (\lambda_1, \lambda_2, ..., \lambda_S)$, where $\sum_{s=1}^{S} \lambda_s = 1$ and each λ_s is divisible by ε. All candidate weights satisfying the above definition constitute the weight set $\mathcal{W}$. Further, We set ε to 0.02 and record the inference results of each test sample under all weights in $\mathcal{W}$. In Table 1, "Batch-wise" refers to the scenario where each test batch is assigned the optimal weight, and the results from all batches are integrated to achieve overall accuracy. The explanation of "Sample-wise" follows a similar logic. In the scenario of dynamic inference, both ensemble learning and model fusion have achieved

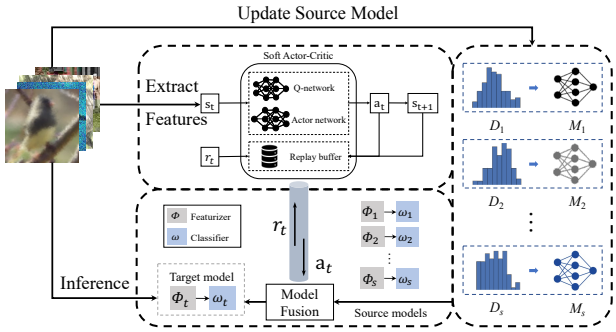


Figure 3: Overview of MFTTA. We employ reinforcement learning for dynamic weight decision-making, and the fused target model is used to infer the test data. The model with the highest correlation with the test batch is selected to update.

performance far exceeding that of existing work. Moreover, the performance of model fusion has significantly improved compared to ensemble learning, indicating that model fusion has the potential to surpass traditional outputs ensemble methods.

4 Method

4.1 Framework Overview

The objective of continuous TTA is to improve the performance of source models in an online manner. Unlabeled test data is provided sequentially, and the model can only access the data at the current moment. Specifically, the test data sequence $\{x_i^{(1)}\}_{i=1}^B \rightarrow \{x_i^{(2)}\}_{i=1}^B \rightarrow \ldots \{x_i^{(t)}\}_{i=1}^B \rightarrow \ldots (B > 0)$ arrives in online form in batches, where t is the timestamp index, B is the number of samples in the test batch. At time t, model M_s needs to predict the target data $\{x_i^{(t)}\}_{i=1}^B$, and adjust θ_s accordingly for future inputs. Each sample is assigned a unique weight. As shown in Fig. 3, for each test image, we first use the pre-trained model to extract features, and then we generate the action $\mathbf{a_t}$ by using the actor-network trained on the soft actor-critic (SAC) algorithm [13]. Then calculate the interpolation weight $\Lambda^{(t)} = (\lambda_1^{(t)}, \lambda_2^{(t)}, ..., \lambda_S^{(t)})$ according to $\mathbf{a_t}$, and implement model fusion according to the Eq. 2. The target model obtained by fusion is used to infer the test data, and the results are used to calculate the reward r_t and return it to the RL agent for update.

4.2 Dynamic Fusion of Source Models

State. Since the test input is a picture, we use a pre-trained model M_F to extract features from the picture as the state, as shown in Eq. 3. Considering the complexity of the source model, the lightweight MobileNetV2 (excluding the linear layer) is used to implement feature extraction. The obtained 1280-dimensional features are used as the dimensions of the state space, thereby avoiding large computing power overhead in the feature extraction process.

$$\mathbf{s_t} = M_F(x_i^{(t)}). \tag{3}$$

Action. We use a continuous action space, with the dimension set to the number of source domains, so the model outputs actions $\mathbf{a_t} = \{a_1, a_2, ..., a_S\}$ in each interaction. We associate the actions output by the RL model with the interpolation weights. The calculation method is shown in the Eq. 4:

$$\Lambda = \{\lambda_1, \lambda_2, ..., \lambda_S\} = \text{Softmax}(\{a_1, a_2, ..., a_S\}). \tag{4}$$

Reward. We adopt Shannon entropy, which is widely used in TTA, as the reward function. Although in deep neural networks, optimizing Shannon entropy will cause the weight decision to fall into a local optimal solution [39], the replay buffer of reinforcement learning can avoid this problem through historical record replay. Therefore, the reward function is shown in Eq. 5, where $\hat{y}_i^{(t)}$ is the output of the target model M_t, β is the scaling factor:

$$r_t = -\beta H(\hat{y}_i^{(t)}). \tag{5}$$

Next, we describe the process of training RL agents and implementing adaptative weight decision-making. First, we need to initialize the policy network parameters θ, replay buffer $\mathcal{R}$, Q network parameters φ_1, φ_2 and target Q network parameters $\varphi_1^{tar}, \varphi_2^{tar}$. The policy network is used to output the mean and variance to construct a continuous probability density function, and the action is sampled from the distribution. The above five networks are all represented by multi-layer perception (MLP) with 2 hidden layers. The policy network replaces the last linear layer with two parallel linear layers (i.e., "multi-head" structure) to output the mean and variance. For the test sample $x_i^{(t)}$, M_F is used to extract features as the state $\mathbf{s_t}$, and then use the policy network $\pi_\theta(\cdot|\mathbf{s_t})$ to select the action $\mathbf{a_t}$, which is converted into interpolation weight according to Eq. 4. The constructed target model M_t is used to infer the test sample $x_i^{(t)}$, and the inference result is calculated according to the formula Eq. 5 to obtain the reward value r_t. Finally, the next state $\mathbf{s_{t+1}}$ and the termination signal d are obtained, and the tuple $(\mathbf{s_t}, \mathbf{a_t}, r_t, \mathbf{s_{t+1}}, d)$ is deposited into the replay buffer $\mathcal{R}$. d is used to determine whether the test set inference is completed and the algorithm terminates.

When the RL model needs to be updated, $|\mathsf{B}|$ records are selected from $\mathcal{R}$ to form the training batch B. Then the gradients of the two Q networks are calculated according to the Eq. 6 ($h \in \{1, 2\}$):

$$\nabla_{\varphi_h} \frac{1}{|\mathsf{B}|} \sum_{(\mathbf{s},\mathbf{a},r,\mathbf{s}',d)\in\mathsf{B}} \left(Q_h(\mathbf{s}, \mathbf{a}; \varphi_h) - y(r, \mathbf{s}', d)\right)^2, \tag{6}$$

where φ_h is the parameter of the h-th Q network, $y(r, \mathbf{s}', d)$ is calculated for each record $(\mathbf{s}, \mathbf{a}, r, \mathbf{s}', d)$ to determine the target component in the Q function, as illustrated in Eq. 7:

$$\begin{aligned} &y(r, \mathbf{s}', d) \\ &= r + \gamma(1-d)\left(\min_{h\in\{1,2\}} Q_h^{tar}(\mathbf{s}', \tilde{\mathbf{a}}'; \varphi_h^{tar}) - \alpha \log \pi_\theta(\tilde{\mathbf{a}}'|\mathbf{s}')\right), \end{aligned} \tag{7}$$

where φ_h^{tar} is the parameter of the h-th target Q network, and $\tilde{\mathbf{a}}'$ is sampled from $\pi_\theta(\cdot|\mathbf{s}')$. Subsequently, the gradient of the policy network is calculated according to Eq. 8:

$$\nabla_\theta \frac{1}{|\mathsf{B}|} \sum_{\mathbf{s}\in\mathsf{B}} \left(\min_{h=1,2} Q_h(\mathbf{s}, \tilde{\mathbf{a}}_\theta(\mathbf{s}); \varphi_h) - \alpha \log \pi_\theta(\tilde{\mathbf{a}}_\theta(\mathbf{s})|\mathbf{s})\right), \tag{8}$$

where $\tilde{\mathbf{a}}_\theta(\mathbf{s})$ is obtained through re-parameterization technique, sampled from $\pi\theta(\cdot|\mathbf{s}')$, with the intention of making it differentiable to the policy network parameters θ. Finally, according to Eq. 9, the updated Q network and the corresponding target Q network are weighted and averaged to serve as the new parameter values for the target Q network ($h \in 1, 2$):

$$\varphi_h^{tar} \leftarrow \rho\varphi_h^{tar} + (1-\rho)\varphi_h. \tag{9}$$

Table 2: Top-1 accuracy on CIFAR-100-C during adaptation. For each single-source TTA method, X+MFTTA outperforms X, indicating that MFTTA offers better generalization performance compared to single-source TTA methods.

	Avg	GN	SN	IN	DB	GB	MB	ZB	Snow	Frost	Fog	Bri	Cont	Elas	Pix	JPEG
Source Worst	48.1	**47.8**	**56.6**	58.7	32.4	**55.2**	40.0	38.6	70.2	**73.1**	28.5	77.8	6.9	57.5	17.8	60.3
Source Best	**64.2**	42.9	50.2	**69.2**	**74.9**	50.6	**72.1**	**76.4**	**75.3**	68.9	**53.9**	**84.8**	**35.9**	**73.6**	**46.6**	**87.9**
Tent Best	74.8	**82.0**	83.1	75.2	85.3	69.3	77.6	81.7	74.8	75.8	68.8	78.1	72.7	65.7	70.0	62.6
Tent+MFTTA	**85.9**	80.7	**85.1**	**76.5**	**94.4**	**79.3**	**89.4**	**92.2**	**86.7**	**87.4**	**78.6**	**94.1**	**90.4**	**83.6**	**90.5**	**79.3**
CoTTA Best	86.0	**79.7**	**84.0**	**72.9**	94.8	72.8	90.2	92.5	86.6	88.3	75.2	94.4	**92.7**	84.8	91.7	**89.5**
CoTTA+MFTTA	**86.5**	78.9	82.4	72.8	**95.8**	**78.7**	**91.6**	**93.9**	**88.0**	**88.7**	**78.8**	**95.6**	92.6	**85.5**	**93.2**	81.3
EATA Best	77.1	76.8	78.4	73.1	82.4	76.6	80.5	85.2	79.1	77.7	72.2	79.1	74.6	73.4	79.2	67.6
EATA+MFTTA	**85.0**	**81.0**	**84.2**	**77.3**	**92.3**	**77.9**	**88.1**	**90.7**	**85.2**	**86.7**	**78.6**	**92.4**	**89.4**	**82.2**	**89.3**	**79.3**
SAR Best	85.6	**80.6**	83.7	73.2	**95.3**	71.9	90.3	92.6	85.4	86.4	75.6	**95.2**	91.5	81.8	91.0	**89.5**
SAR+MFTTA	**86.4**	80.0	**84.6**	**76.4**	95.0	**78.3**	**90.4**	**92.8**	**87.2**	**88.4**	**79.9**	94.9	**91.5**	**84.4**	**91.6**	80.0

4.3 Adaptive Model Selection

Similar to single-source domain TTA methods, we also implement updates for source models. At timestep t, the interpolation weights for the accumulated batch are averaged to obtain the average weight $\overline{\lambda}^{(t)} = \{\overline{\lambda}_1^{(t)}, \overline{\lambda}_2^{(t)}, ..., \overline{\lambda}_S^{(t)}\}$. The source model selection method, as shown in Eq. 10, indicates that the current test batch distribution is most closely related to the source domain model M_k:

$$k = \arg\max_{1 \leq s \leq S} \overline{\lambda}^{(t)}. \tag{10}$$

For the selected source model M_k, single-source domain TTA methods are employed to adapt M_k to real-time data distribution.

5 Experiments

5.1 Experimental Settings

Datasets and models. We conduct experiments on three OOD datasets: CIFAR-100-C [14], ImageNet-C [14] and DomainNet-126 [32]. CIFAR-100-C and ImageNet-C have undergone 15 different types of corruption, each with five different intensity levels. The corruption is divided into four categories, namely "Noise, Blur, Weather, Digital", where Noise includes "Gaussian Noise, Shot Noise, Impulse Noise", Blur includes "Defocus Blur, Glass Blur, Motion Blur, Zoom Blur", Weather includes "Frost, Snow, Fog, Brightness", and Digital includes "Contrast, Elastic Transform, Pixelate, JPEG Compression". Abbreviations are used in the experimental results, such as abbreviating Gaussian Noise as GN. DomainNet-126 is a subset of DomainNet [30]. Since the label noise of DomainNet is large, DomainNet-126 selects 126 categories and 4 domains, namely Clipart, Painting, Real, and Sketch [32]. Following the settings of RobustBench benchmark [8], the source model for CIFAR-100-C experiments is initialized with a ResNeXt-29 model pre-trained on CIFAR-100. The ImageNet-C and DomainNet-126 experiments utilize a ResNet-50 model pre-trained on ImageNet.

Baseline Methods. Single-source TTA methods are used as experimental benchmarks, including Tent [39], CoTTA [42], EATA [26], and SAR [27]. To assess the adaptability of models, we adopt settings similar to DECISION [1], applying each source model individually to the test data of a specific domain to derive results for the X Best. "X" represents the name of the single-source TTA method, with the terms respectively indicating the scenarios of highest performance when adapting source models. We also present scenarios using only the original collection of source models without any adaptation, labeled accordingly as Source Best and Source Worst. All single-source domain methods "X" are extended to a multi-source domain environment, with the multi-source collaborative method referred to as "X+MFTTA".

Implementation details. For the reinforcement learning models, the learning rates for the policy network and the Q network are set at 0.0001. The reward discount factor γ is set to 0.9, the exploration rate parameter α to 0.2, the update interval $\mathsf{T_R}$ to 15, and the number of updates N to 15. In the CIFAR-100-C experiments, the scaling factor β is set to 1, the source model update interval T_S to 200, and the reinforcement learning update batch size $|\mathsf{B}|$ to 1024. For the ImageNet-C and DomainNet-126 experiments, the scaling factor β is set to 5, the source model update interval T_S to 50, and the reinforcement learning update batch size $|\mathsf{B}|$ to 50. For CIFAR-100-C and ImageNet-C, we selected Gaussian Noise, Glass Blur, Snow, and Jpeg Compression with a severity level of 1 to train source models. Under severity level 5, different domains are continuously used to adapt the source models. For DomainNet-126, we select two of the domains as source domains and the other two as target domains.

5.2 Experimental Results

Experiments on CIFAR-100-C. Table 2 shows the experimental results of CIFAR-100-C. Even for the highest-performing source model, its inference accuracy lags far behind the single-source domain TTA method because it does not perform test phase adaptation. For the single-source domain TTA method "X Best", it is observed that CoTTA and SAR are significantly improved compared to Source Best, while Tent and EATA are slightly worse. Compared with the best single-source method, the "X+MFTTA" method proposed in this paper can significantly improve the average inference accuracy. This shows that "X+MFTTA" can use appropriate weights to achieve dynamic model fusion when inputting test data, thereby achieving better performance than the best single-source domain method. However, while there are substantial improvements for Tent and EATA, there is only a 0.5% improvement for CoTTA and only a 0.8% improvement for SAR. Furthermore, it can be observed that the average inference accuracy of Tent and EATA after adaptation to the 15 test domains is significantly lower than that of CoTTA and SAR. Among the dynamic model fusion methods in this paper, the performance of Tent+MFTTA and EATA+MFTTA is equivalent to that of CoTTA+MFTTA and SAR+MFTTA, both at around 86%, indicating that MFTTA can bridge the gap between single-source TTA methods and can be generalized to various single-source TTA scenarios.

Experiments on ImageNet-C. Table 3 shows the experimental results of ImageNet-C. Compared with the CIFAR-100-C experiment, when the input complexity increases, the gap between the source model inference and the single-source domain TTA method fur-

Table 3: Top-1 accuracy on ImageNet-C during adaptation. For each single-source TTA method, the performance of X+MFTTA surpasses that of X, indicating that MFTTA offers better generalization performance compared to single-source TTA methods.

	Avg	GN	SN	IN	DB	GB	MB	ZB	Snow	Frost	Fog	Bri	Cont	Elas	Pix	JPEG
Source Worst	18.8	0.6	1.0	0.4	**5.7**	**7.7**	**10.8**	**22.0**	14.3	16.1	**31.2**	67.9	**6.0**	**23.4**	32.8	42.4
Source Best	**25.2**	**10.2**	**14.6**	**10.4**	4.1	5.6	8.6	15.5	**22.9**	**37.0**	17.5	**80.8**	0.5	17.4	**49.5**	**83.6**
Tent Best	57.0	36.7	47.3	49.2	29.8	34.0	43.7	63.3	50.2	53.5	66.2	87.7	51.9	72.4	87.5	81.9
Tent+MFTTA	**64.4**	**42.0**	**50.0**	**50.8**	**33.4**	**37.3**	**53.1**	**68.5**	**74.9**	**63.9**	**78.6**	**93.5**	**63.8**	**76.8**	**90.2**	**89.7**
CoTTA Best	56.0	33.2	40.6	42.5	31.0	**34.4**	48.1	62.1	54.5	56.8	67.2	82.5	60.7	71.7	79.5	74.6
CoTTA+MFTTA	**63.0**	**40.7**	**46.2**	**46.2**	**31.3**	30.6	**50.3**	**66.0**	**73.5**	**64.1**	**78.5**	**93.6**	**68.0**	**77.6**	**90.0**	**88.4**
EATA Best	67.2	45.2	**60.7**	**60.6**	38.3	43.2	57.0	72.2	61.3	61.9	78.4	92.5	**71.6**	**80.6**	**94.5**	90.4
EATA+MFTTA	**69.0**	**47.0**	59.3	59.3	**40.7**	**45.0**	**61.3**	**74.1**	**73.4**	**65.6**	**80.7**	**94.1**	69.7	79.7	93.6	**92.0**
SAR Best	58.5	38.0	41.3	38.3	28.6	28.6	44.7	62.0	55.2	55.4	74.4	**94.1**	61.1	76.2	**93.7**	85.6
SAR+MFTTA	**64.6**	**42.1**	**50.1**	**51.6**	**32.0**	**35.0**	**51.8**	**68.4**	**75.8**	**64.3**	**78.5**	93.7	**66.1**	**77.7**	90.7	**90.7**

Table 4: Comparison of our method and SFDA methods on ImageNet-C. Our method achieves the highest average accuracy.

	Avg	GN	SN	IN	DB	GB	MB	ZB	Snow	Frost	Fog	Bri	Cont	Elas	Pix	JPEG
DECISION-17	54.6	37.2	39.8	39.4	26	22.5	39.4	53.5	59.5	56.6	73.6	90.4	54.7	63.6	75.9	87.1
DECISION-50	55.5	30	36.8	33.2	28.3	23.8	41	58.4	68.5	61.5	76.4	93.5	45.8	68.6	77.9	88.9
Shen-8	66.5	**57.3**	**59.8**	56.5	**44.2**	36.3	54.8	72.2	**84.6**	**70.3**	**81.0**	**94.8**	50.3	60.6	85.4	90
EAAF-17	51.2	27.6	31.4	28.6	21.6	23.0	35.4	52.5	70.7	52.9	69.7	93.2	35.3	57.3	78.9	90.2
EAAF-50	56.7	35.1	39.3	38.0	25.2	26.0	40.0	58.6	78.1	56.3	75.1	94.4	46.4	64.4	82.5	91.7
Ours Best	**69.0**	47.0	59.3	**59.3**	40.7	**45.0**	**61.3**	**74.1**	73.4	65.6	80.7	94.1	**69.7**	**79.7**	**93.6**	**92**

ther widens. For Tent, CoTTA and SAR, the dynamic model fusion method can improve the inference accuracy by at least 10%, and EATA has an accuracy improvement of 1.8%. Compared with CIFAR-100-C, the dynamic model fusion method has significant performance improvements over the single-source domain TTA method. In addition, although Source Worst corresponds to the Glass Blur source model, under the 4-type single source domain TTA method, the corresponding X Best source domains are all Glass Blur, indicating that the source domain with the worst initial performance is not necessarily the worst. On the contrary, it may become the source model with the best performance under TTA update.

Experiments on DomainNet-126. Table 5 presents the experimental results on the DomainNet-126 dataset. Across all domain configurations, Source Best consistently outperforms Source Worst, demonstrating a clear performance gap. Single-source TTA methods offer only limited improvements over the Source Best, and in some cases, may even degrade the performance of the source models. For example, in the "P+S→C+R" setting, SAR Best slightly underperforms Source Best, indicating that the SAR method fails to adapt the source model to the evolving data distribution and may even impair its original performance. Similar observations hold for EATA Best and SAR Best in the "R+S→C+P" scenario. In contrast, the inference accuracy of all X+MFTTA methods is consistently higher than that of the corresponding X Best, showing none of the aforementioned issues. Since X+MFTTA also involves single-source TTA updates, this result suggests that dynamic model fusion can still stably improve inference accuracy on test data, even in scenarios where the source model is prone to degradation.

Experiments of SFDA methods. Facing the research gap of multi-source model collaboration in continuous TTA scenarios, we introduce SFDA methods to verify the effectiveness of our proposed method. We select three research works, DECISION[1], Shen[33], and EAAF[28], for comparison. We treat each batch as a test set and reset the source model and hyper-parameters at each timestep. Table 4 shows the experimental results, among which Ours best represents EATA+MFTTA. Furthermore, a larger impact of batch size was observed for DECISION and EAAF methods. It is first observed that Ours best is the method with the highest average performance, outperforming the next-best method by 2.5%. For DECISION, a configuration with a batch size of 50 results in nearly 1% performance improvement over 17, while for the EAAF method, a batch size of 50 results in approximately 5.5% performance improvement. Therefore, it can be inferred that larger batch size may be more beneficial for the SFDA method when the size of the test set is small. In addition, the SFDA method needs to use test data to fine-tune the source model at each moment, which introduces computational overhead and makes performance sensitive to the test batch size and the number of adaptation iterations. In contrast, our approach maintains high efficiency without relying on consistent test-time fine-tuning.

Table 5: Top-1 accuracy on DomainNet-126 during adaptation. For each single-source TTA method, the performance of X+MFTTA surpasses that of X, indicating that MFTTA offers better generalization performance compared to single-source TTA methods.

	P+S→C+R	C+R→P+S	C+P→R+S	R+S→C+P
Source Worst	57.90	49.72	57.11	54.71
Source Best	**67.48**	**58.36**	**66.32**	**60.59**
Tent Best	66.97	60.97	67.61	61.29
Tent+MFTTA	**68.94**	**63.24**	**69.34**	**65.20**
CoTTA Best	69.75	62.37	68.75	63.50
CoTTA+MFTTA	**70.63**	**63.93**	**70.04**	**66.00**
EATA best	67.72	60.51	68.09	60.57
EATA+MFTTA	**69.25**	**62.72**	**69.16**	**64.71**
SAR Best	66.62	60.15	67.05	60.32
SAR+MFTTA	**69.38**	**63.36**	**69.77**	**65.28**

Experiments of forgetting on CIFAR-100-C. We investigate the ability of our method to prevent ID forgetting during adaptation. We measure the anti-forgetting ability by comparing the classification accuracy of the model before and after adaptation on the corresponding source domain data. Fig. 4 shows that Tent and EATA have obvious forgetting phenomena. CoTTA is relatively slight in the early stage

Table 6: Top-1 accuracy of weight decision strategies on ImageNet-C. Our proposed RL-based weight decision strategy achieves the highest average accuracy across all configurations.

X+MFTTA	Avg	GN	SN	IN	DB	GB	MB	ZB	Snow	Frost	Fog	Bright	Contrast	Elastic	Pixel	JPEG
Tent (Rand)	60.86	39.22	45.16	44.72	30.76	31.56	47.90	65.62	65.76	62.52	76.70	93.44	61.66	73.90	85.96	87.98
Tent (Avg)	61.93	41.32	47.98	47.66	31.38	32.32	47.96	66.32	66.58	63.60	77.70	**93.62**	63.44	74.50	85.86	88.76
Tent (RL)	**64.43**	**42.00**	**49.96**	**50.80**	**33.44**	**37.28**	**53.14**	**68.50**	**74.86**	**63.92**	**78.58**	93.52	**63.80**	**76.84**	**90.20**	**89.66**
CoTTA (Rand)	59.98	37.82	41.96	40.24	28.70	28.12	47.14	63.78	66.12	62.68	77.86	93.78	63.34	74.14	87.06	86.96
CoTTA (Avg)	61.39	40.38	44.82	43.46	30.52	28.92	48.10	64.86	67.38	63.44	**78.48**	**93.98**	66.60	74.16	87.76	88.00
CoTTA (RL)	**63.00**	**40.68**	**46.16**	**46.16**	**31.34**	**30.58**	**50.32**	**65.98**	**73.50**	**64.12**	78.46	93.64	**67.98**	**77.58**	**90.04**	**88.42**
EATA (Rand)	64.21	41.56	52.22	54.28	32.72	37.84	52.56	68.34	67.20	64.36	78.44	93.18	67.78	75.86	87.36	89.42
EATA (Avg)	65.32	43.94	55.42	57.54	33.38	38.58	52.94	69.10	68.30	64.78	78.56	93.56	69.24	76.30	87.80	90.30
EATA (RL)	**69.03**	**46.96**	**59.30**	**59.30**	**40.68**	**45.02**	**61.26**	**74.14**	**73.36**	**65.56**	**80.70**	**94.12**	**69.70**	**79.70**	**93.62**	**92.04**
SAR (Rand)	61.39	38.58	45.54	45.34	30.42	31.92	48.28	66.04	66.98	63.04	76.78	93.74	63.14	75.42	86.60	88.98
SAR (Avg)	62.64	41.42	48.28	48.68	31.16	32.36	48.54	67.34	67.70	**64.44**	77.84	**94.12**	65.14	75.98	87.12	89.52
SAR (RL)	**64.56**	**42.10**	**50.12**	**51.60**	**32.00**	**34.98**	**51.78**	**68.44**	**75.76**	64.30	**78.50**	93.66	**66.12**	**77.66**	**90.70**	**90.74**

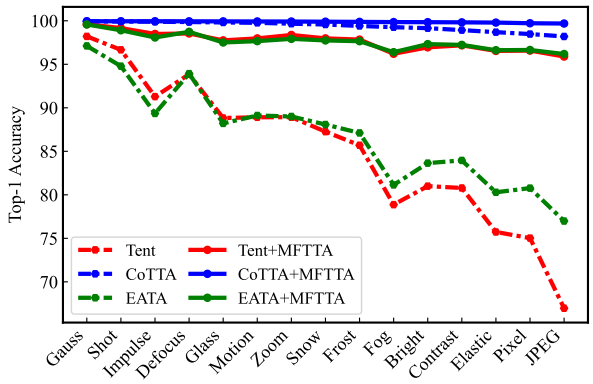


Figure 4: Results of ID forgetting on CIFAR-100-C. By integrating single-source methods with MFTTA, source models can better retain source knowledge during adaptation.

and also declines slightly in the later stage. In comparison, MFTTA can significantly alleviate the forgetting problem.

Ablation studies. To evaluate the source model selection strategies, we considered four ablation approaches: selecting the source model with the highest weight (Most Corr.), the one with the lowest weight (Least Corr.), a randomly selected source model (Random), and using all source models (All) for updating. Except for the choice of source models, all other experimental settings remain consistent with those described earlier. Table 7 shows that updating only the source model with the highest correlation to the target domain yields the best test accuracy, outperforming the second-best method by an average of 2.77% in Tent+MFTTA, CoTTA+MFTTA, and SAR+MFTTA. However, in the case of EATA+MFTTA, updating all source models achieves the best performance, surpassing the highest-correlation model strategy by 2.51% in accuracy. Moreover, across all four groups of experiments, updating the source model with the lowest weight consistently results in the worst performance.

In addition to the source model selection, we also investigate the impact of different weight decision mechanisms on inference accuracy. Specifically, the following methods are compared: randomly generating interpolation weights (Random), equal weights assigned to each source domain (Avg), and reinforcement learning-based weight decision (RL). Each of these three weighting strategies is applied to the "X+MFTTA" framework variants. For both the random and reinforcement learning-based weighting methods, the source model update strategy selects the source model with the highest average weight. In contrast, for the average weighting method, a randomly selected source model is used for updating. The results

Table 7: Top-1 accuracy of source model selection strategies on ImageNet-C. Updating the most relevant source model is effective in most cases.

X+MFTTA	All	Random	Least Corr.	Most Corr.
Tent	63.00	62.57	62.24	**64.43**
CoTTA	61.80	61.84	61.55	**63.00**
EATA	**71.54**	65.83	64.72	69.03
SAR	58.85	57.23	57.05	**64.56**

on the ImageNet-C dataset are shown in Table 6. Among the three approaches, the proposed RL-based weighting strategy achieves the highest accuracy, followed by average weighting, while random weighting yields the lowest performance. For the majority of target domains, the RL-based method demonstrates the best performance. Even in relatively weaker cases such as the Bright domain, its performance gap with the best method remains within 5%. Furthermore, except for the EATA+MFTTA configuration, both average and random weighting outperform the single-source TTA baseline in the other three settings. This indicates that even without explicit weight learning, random weighting is sufficient to surpass single-source TTA methods.

6 Conclusion

In this paper, we propose MFTTA, a novel TTA algorithm that effectively integrates multiple source models through parameter-level fusion while substantially mitigating the forgetting problem in long-term adaptation. By leveraging an RL-based fusion strategy and a correlation-aware update mechanism, MFTTA constructs adaptive target models that generalize well under distribution shifts. Experimental results on standard OOD benchmarks confirm that MFTTA not only outperforms traditional single-source adaptation approaches but also achieves superior performance compared to existing multi-source TTA methods.

Acknowledgements

National Key Research and Development Project of China (Grant No. 2023YFF0905502), National Natural Science Foundation of China (Grant No. 92467204, 62472249 and 62402264), and Shenzhen Science and Technology Program (Grant No. JCYJ20220818101014030 and KJZD20240903102300001).

References

[1] S. M. Ahmed, D. S. Raychaudhuri, S. Paul, S. Oymak, and A. K. Roy-Chowdhury. Unsupervised multi-source domain adaptation without access to source data. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10103–10112, 2021.

[2] S. Ainsworth, J. Hayase, and S. Srinivasa. Git re-basin: Merging models modulo permutation symmetries. In *The Eleventh International Conference on Learning Representations*, 2022.

[3] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz. Invariant risk minimization. *CoRR*, abs/1907.02893, 2019.

[4] D. Arpit, H. Wang, Y. Zhou, and C. Xiong. Ensemble of averages: Improving model selection and boosting performance in domain generalization. *Advances in Neural Information Processing Systems*, 35: 8265–8277, 2022.

[5] J. Cha, S. Chun, K. Lee, H.-C. Cho, S. Park, Y. Lee, and S. Park. Swad: Domain generalization by seeking flat minima. *Advances in Neural Information Processing Systems*, 34:22405–22418, 2021.

[6] A. Choromanska, M. Henaff, M. Mathieu, G. Ben Arous, and Y. LeCun. The Loss Surfaces of Multilayer Networks. In *Proceedings of the Eighteenth International Conference on Artificial Intelligence and Statistics*, volume 38 of *Proceedings of Machine Learning Research*, pages 192–204, San Diego, California, USA, 09–12 May 2015. PMLR.

[7] L. Choshen, E. Venezian, N. Slonim, and Y. Katz. Fusing finetuned models for better pretraining. *arXiv preprint arXiv:2204.03044*, 2022.

[8] F. Croce, M. Andriushchenko, V. Sehwag, E. Debenedetti, N. Flammarion, M. Chiang, P. Mittal, and M. Hein. Robustbench: a standardized adversarial robustness benchmark. *Advances in Neural Information Processing Systems*, 2021.

[9] R. Entezari, H. Sedghi, O. Saukh, and B. Neyshabur. The role of permutation invariance in linear mode connectivity of neural networks. In *International Conference on Learning Representations*, 2021.

[10] P. Foret, A. Kleiner, H. Mobahi, and B. Neyshabur. Sharpness-aware minimization for efficiently improving generalization. In *International Conference on Learning Representations*, 2020.

[11] T. Gong, J. Jeong, T. Kim, Y. Kim, J. Shin, and S.-J. Lee. Note: Robust continual test-time adaptation against temporal correlation. In *Advances in Neural Information Processing Systems*, volume 35, pages 27253–27266. Curran Associates, Inc., 2022.

[12] I. Gulrajani and D. Lopez-Paz. In search of lost domain generalization. In *International Conference on Learning Representations*, 2020.

[13] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1861–1870. PMLR, 10–15 Jul 2018.

[14] D. Hendrycks and T. Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations*, 2018.

[15] X. Hu, M. G. Uzunbas, S. Chen, R. Wang, A. Shah, R. Nevatia, and S. Lim. Mixnorm: Test-time adaptation through online normalization estimation. *CoRR*, abs/2110.11478, 2021.

[16] P. Izmailov, D. Podoprikhin, T. Garipov, D. Vetrov, and A. G. Wilson. Averaging weights leads to wider optima and better generalization. *arXiv preprint arXiv:1803.05407*, 2018.

[17] X. Jin, X. Ren, D. Preotiuc-Pietro, and P. Cheng. Dataless knowledge fusion by merging weights of language models. In *11th International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.

[18] R. Kuditipudi, X. Wang, H. Lee, Y. Zhang, Z. Li, W. Hu, R. Ge, and S. Arora. Explaining landscape connectivity of low-cost solutions for multilayer nets. In *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019.

[19] D. Li, Y. Yang, Y.-Z. Song, and T. Hospedales. Learning to generalize: Meta-learning for domain generalization. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.

[20] J. Liang, D. Hu, Y. Wang, R. He, and J. Feng. Source data-absent unsupervised domain adaptation through hypothesis transfer and labeling transfer. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(11):8602–8617, 2022.

[21] J. Liang, R. He, and T. Tan. A comprehensive survey on test-time adaptation under distribution shifts. *International Journal of Computer Vision*, pages 1–34, 2024.

[22] Y. Liu, P. Kothari, B. van Delft, B. Bellot-Gurlet, T. Mordan, and A. Alahi. Ttt++: When does self-supervised test-time training fail or thrive? In M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 21808–21820. Curran Associates, Inc., 2021.

[23] M. Mancini, S. R. Bulò, B. Caputo, and E. Ricci. Best sources forward: Domain generalization through source-specific nets. In *2018 25th IEEE International Conference on Image Processing (ICIP)*, pages 1353–1357, 2018.

[24] M. S. Matena and C. A. Raffel. Merging models with fisher-weighted averaging. In *Advances in Neural Information Processing Systems*, volume 35, pages 17703–17716. Curran Associates, Inc., 2022.

[25] A. R. Nelakurthi, R. Maciejewski, and J. He. Source free domain adaptation using an off-the-shelf classifier. In *2018 IEEE International Conference on Big Data (Big Data)*, pages 140–145, 2018.

[26] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, and M. Tan. Efficient test-time model adaptation without forgetting. In *International conference on machine learning*, pages 16888–16905. PMLR, 2022.

[27] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, and M. Tan. Towards stable test-time adaptation in dynamic wild world. In *The Eleventh International Conference on Learning Representations*, 2023.

[28] J. Pei, A. Men, Y. Liu, X. Zhuang, and Q. Chen. Evidential multi-source-free unsupervised domain adaptation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pages 1–18, 2024.

[29] Z. Pei, Z. Cao, M. Long, and J. Wang. Multi-adversarial domain adaptation. In *Proceedings of the AAAI conference on artificial intelligence*, volume 32, 2018.

[30] X. Peng, Q. Bai, X. Xia, Z. Huang, K. Saenko, and B. Wang. Moment matching for multi-source domain adaptation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 1406–1415, 2019.

[31] A. Ramé, K. Ahuja, J. Zhang, M. Cord, L. Bottou, and D. Lopez-Paz. Model ratatouille: Recycling diverse models for out-of-distribution generalization. In *International Conference on Machine Learning*, pages 28656–28679. PMLR, 2023.

[32] K. Saito, D. Kim, S. Sclaroff, T. Darrell, and K. Saenko. Semi-supervised domain adaptation via minimax entropy. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 8050–8058, 2019.

[33] M. Shen, Y. Bu, and G. W. Wornell. On balancing bias and variance in unsupervised multi-source-free domain adaptation. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 30976–30991. PMLR, 23–29 Jul 2023.

[34] Z. Shen, J. Liu, Y. He, X. Zhang, R. Xu, H. Yu, and P. Cui. Towards out-of-distribution generalization: A survey. *CoRR*, abs/2108.13624, 2021.

[35] S. P. Singh and M. Jaggi. Model fusion via optimal transport. *Advances in Neural Information Processing Systems*, 33:22045–22055, 2020.

[36] G. Stoica, D. Bolya, J. Bjorner, P. Ramesh, T. Hearn, and J. Hoffman. Zipit! merging models from different tasks without training. In *International Conference on Learning Representations*. International Conference on Learning Representations, 2024.

[37] B. Sun and K. Saenko. Deep coral: Correlation alignment for deep domain adaptation. In *Computer Vision–ECCV 2016 Workshops: Amsterdam, The Netherlands, October 8-10 and 15-16, 2016, Proceedings, Part III 14*, pages 443–450. Springer, 2016.

[38] V. Vapnik and V. Vapnik. Statistical learning theory wiley. *New York*, 1 (624):2, 1998.

[39] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell. Tent: Fully test-time adaptation by entropy minimization. In *International Conference on Learning Representations*, 2020.

[40] J. Wang, C. Lan, C. Liu, Y. Ouyang, T. Qin, W. Lu, Y. Chen, W. Zeng, and P. S. Yu. Generalizing to unseen domains: A survey on domain generalization. *IEEE Transactions on Knowledge and Data Engineering*, 35(8):8052–8072, 2023.

[41] M. Wang and W. Deng. Deep visual domain adaptation: A survey. *Neurocomputing*, 312:135–153, 2018.

[42] Q. Wang, O. Fink, L. Van Gool, and D. Dai. Continual test-time domain adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7201–7211, 2022.

[43] M. Wortsman, G. Ilharco, S. Y. Gadre, R. Roelofs, R. Gontijo-Lopes, A. S. Morcos, H. Namkoong, A. Farhadi, Y. Carmon, S. Kornblith, and L. Schmidt. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 23965–23998. PMLR, 17–23 Jul 2022.

[44] K. Zhou, Y. Yang, Y. Qiao, and T. Xiang. Domain adaptive ensemble learning. *IEEE Transactions on Image Processing*, 30:8008–8018, 2021.